

Expt. No. : 10

Date :

HANDLING MISSING DATA

AIM

To write a python program to handle the missing data using pandas.

SOFTWARE REQUIRED

Python 3.14

THEORY

Pandas is a powerful data manipulation library in Python that provides easy-to-use functions for handling missing data. Here are some key concepts and methods for dealing with missing values in Pandas:

1. Identifying Missing Data

Pandas uses NaN (Not a Number) to represent missing values. You can identify missing data using functions like:

- `isna()` or `isnull()`: These functions return a boolean DataFrame indicating the presence of missing values.
- `notna()` or `notnull()`: These functions return a boolean DataFrame indicating non-missing values.

2. Handling Missing Data

Once identified, there are several strategies to handle missing data:

- **Removing Missing Data:**
 - `dropna()`: This function removes rows or columns that contain missing values. You can specify the axis (0 for rows, 1 for columns) and the threshold for minimum non-null values to keep.
- **Filling Missing Data:**
 - `fillna()`: This function allows you to replace missing values with a specified value, such as the mean, median, mode, or a fixed value. You can also use forward fill (`ffill`) or backward fill (`bfill`) to propagate the next or previous values, respectively.

3. Examples of Handling Missing Data

Here are some common operations for handling missing data in Pandas:

- **Remove rows with any missing values:**
`df.dropna()`
- **Fill missing values with the mean of the column:**
`df.fillna(df.mean())`
- **Forward fill missing values:**
`df.fillna(method='ffill')`

PROBLEM

1. Build a Python program using Pandas to identify columns containing missing values and remove the columns with excessive missing data.

DATASET

```
data = {
    "Student_ID": [101, 102, 103, 104, 105, 106, 107, 108, 109, 110],
    "Name": ["Arun", "Bala", "Charan", "Divya", "Ezhil",
            "Farah", "Gokul", "Hari", "Indhu", "Jaya"],
    "Age": [20, 21, 20, 22, 21, np.nan, 20, 21, 20, 22],
    "Attendance": [92, 85, np.nan, 78, 88, 95, np.nan, 82, 90, 76],
    "Maths": [85, 76, 91, np.nan, 69, 88, 74, np.nan, 95, 68],
    "Physics": [78, np.nan, 89, 72, 75, 92, np.nan, 85, 94, 70],
    "Programming": [88, 82, 95, 80, np.nan, 90, 78, 87, np.nan, 75]
}
```

2. Build a Python program to handle the missing power factor measurements using the backward-fill method and display the cleaned dataset.

DATASET

```
data = {Time": ["09:00", "10:00", "11:00", "12:00", "13:00", "14:00", "15:00", "16:00",
               "17:00", "18:00"],
        "Voltage (V)": [230, 232, np.nan, 235, 238, np.nan, 240, 237, 235, 233],
        "Current (A)": [12.5, np.nan, 13.2, 13.8, 14.1, 14.5, np.nan, 15.2, 14.8, 14.0],
        "Power (kW)": [2.75, 2.81, np.nan, 3.10, 3.25, 3.38, np.nan, 3.50, 3.40, 3.26],
        "Temperature (°C)": [32, 34, 36, np.nan, 40, 42, 43, 45, np.nan, 44],
        "Power Factor": [0.92, 0.91, 0.90, 0.89, np.nan, 0.87, 0.86, 0.85, 0.88, np.nan]
}
```

3. Given two arrays of integers, find which elements in the second array are missing from the first array.

ALGORITHM

STEP 1: Start the program.

STEP 2: Import pandas as pd

STEP 3: Get the elements in the dictionary data

STEP 4: Write the data in dataframe

STEP 5: Identify missing values ,by the representations such as 'NaN' , 'None' ,'-999'

STEP 6: Replace missing values with a specified default value using fillna()function

STEP 7: Drop rows or columns conataining missing values using pandas function dropna()

STEP 8: Replace missing values with a calculated value such as mean,median ,mode

STEP 9: Print the modified dataframes

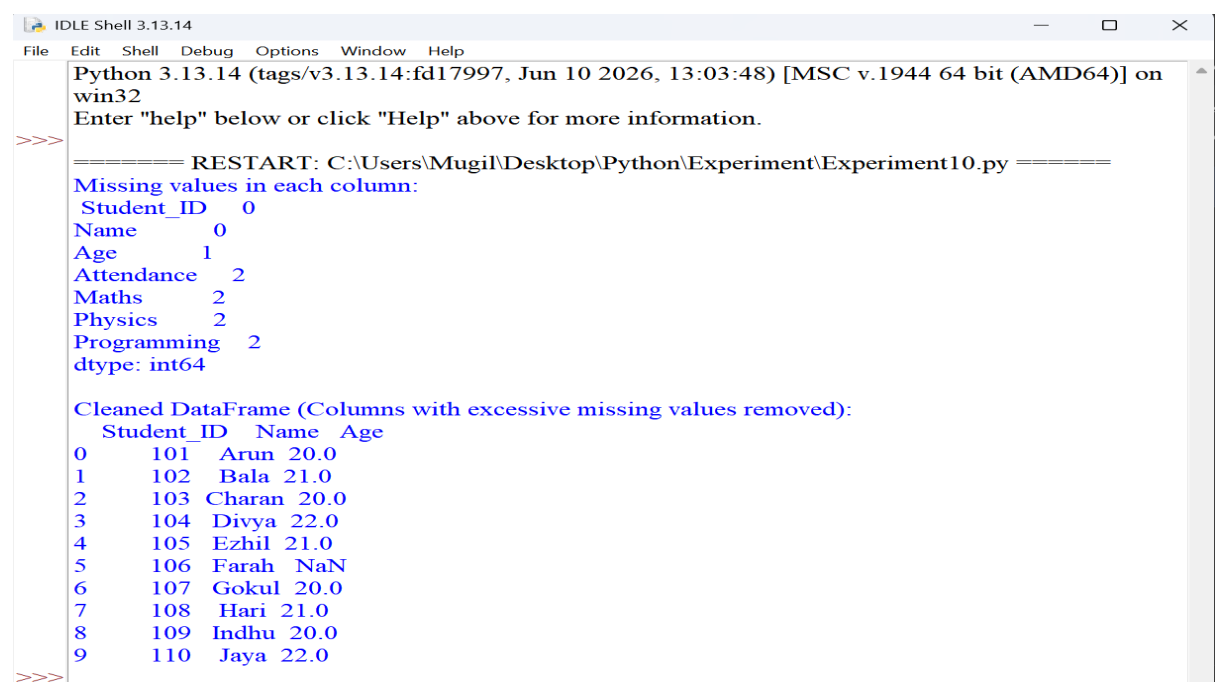
STEP 10: Terminate the program.

PROGRAM CODING

```

import pandas as pd
import numpy as np
data = {
    "Student_ID": [101, 102, 103, 104, 105, 106, 107, 108, 109, 110],
    "Name": ["Arun", "Bala", "Charan", "Divya", "Ezhil", "Farah",
"Gokul", "Hari", "Indhu", "Jaya"],
    "Age": [20, 21, 20, 22, 21, np.nan, 20, 21, 20, 22],
    "Attendance": [92, 85, np.nan, 78, 88, 95, np.nan, 82, 90, 76],
    "Maths": [85, 76, 91, np.nan, 69, 88, 74, np.nan, 95, 68],
    "Physics": [78, np.nan, 89, 72, 75, 92, np.nan, 85, 94, 70],
    "Programming": [88, 82, 95, 80, np.nan, 90, 78, 87, np.nan, 75]
}
df = pd.DataFrame(data)
print("Missing values in each column:\n", df.isnull().sum())
threshold = len(df) - 1
cleaned_df = df.dropna(thresh=threshold, axis=1)
print("\nCleaned DataFrame (Columns with excessive missing values
removed):\n", cleaned_df)

```

OUTCOMES OF PROGRAM CODING


```

IDLE Shell 3.13.14
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\Mugil\Desktop\Python\Experiment\Experiment10.py =====
Missing values in each column:
Student_ID    0
Name          0
Age           1
Attendance    2
Maths         2
Physics       2
Programming   2
dtype: int64

Cleaned DataFrame (Columns with excessive missing values removed):
  Student_ID  Name  Age
0        101  Arun  20.0
1        102  Bala  21.0
2        103  Charan  20.0
3        104  Divya  22.0
4        105  Ezhil  21.0
5        106  Farah  NaN
6        107  Gokul  20.0
7        108  Hari  21.0
8        109  Indhu  20.0
9        110  Jaya  22.0
>>>

```

PROBLEM CODING

```

import pandas as pd
import numpy as np
data = {
    "Time": ["09:00", "10:00", "11:00", "12:00", "13:00", "14:00", "15:00",
    "16:00", "17:00", "18:00"],
    "Voltage (V)": [230, 232, np.nan, 235, 238, np.nan, 240, 237, 235, 233],
    "Current (A)": [12.5, np.nan, 13.2, 13.8, 14.1, 14.5, np.nan, 15.2, 14.8,
    14.0],
    "Power (kW)": [2.75, 2.81, np.nan, 3.10, 3.25, 3.38, np.nan, 3.50, 3.40,
    3.26],
    "Temperature (°C)": [32, 34, 36, np.nan, 40, 42, 43, 45, np.nan, 44],
    "Power Factor": [0.92, 0.91, 0.90, 0.89, np.nan, 0.87, 0.86, 0.85, 0.88,
    np.nan]
}
df = pd.DataFrame(data)
df["Power Factor"] = df["Power Factor"].bfill()
print("Cleaned Dataset:\n", df)

```

OUTCOMES OF PROGRAM CODING

```

IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

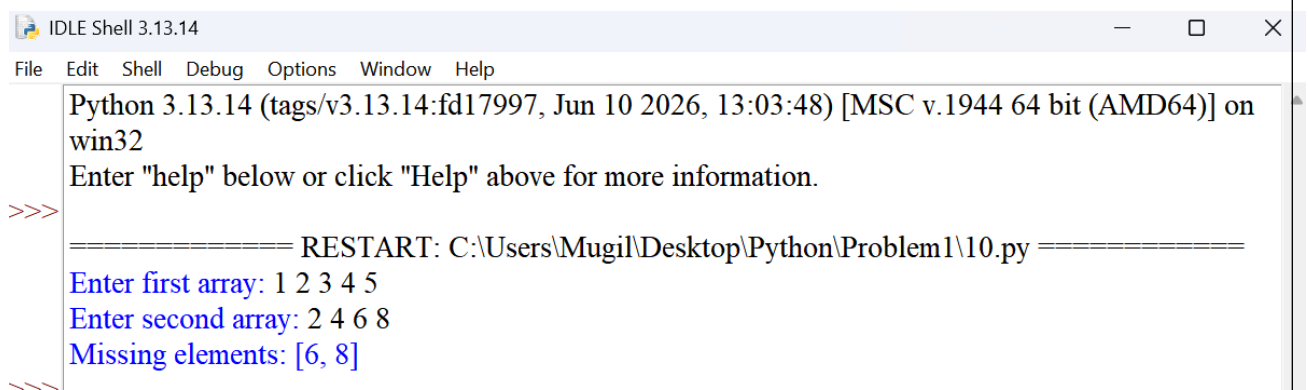
>>>
===== RESTART: C:\Users\Mugil\Desktop\Python\Problem\experiment10.py =====
Cleaned Dataset:
   Time Voltage (V) Current (A) Power (kW) Temperature (°C) Power Factor
0 09:00      230.0      12.5      2.75         32.0         0.92
1 10:00      232.0       NaN      2.81         34.0         0.91
2 11:00       NaN      13.2       NaN         36.0         0.90
3 12:00      235.0      13.8      3.10         NaN         0.89
4 13:00      238.0      14.1      3.25         40.0         0.87
5 14:00       NaN      14.5      3.38         42.0         0.87
6 15:00      240.0       NaN       NaN         43.0         0.86
7 16:00      237.0      15.2      3.50         45.0         0.85
8 17:00      235.0      14.8      3.40         NaN         0.88
9 18:00      233.0      14.0      3.26         44.0         NaN
>>>

```

PROBLEM CODING

```
a = list(map(int, input("Enter first array: ").split()))
b = list(map(int, input("Enter second array: ").split()))
missing = []
for x in b:
    if x not in a:
        missing.append(x)
print("Missing elements:", missing)
```

OUTCOMES OF PROGRAM CODING



```
IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\Mugil\Desktop\Python\Problem1\10.py =====
Enter first array: 1 2 3 4 5
Enter second array: 2 4 6 8
Missing elements: [6, 8]
```

VIVA QUESTION

1. What is the significance of handling missing data in a dataset?

Handling missing data is crucial because it ensures the integrity of the dataset, allows for accurate analysis, and helps prevent bias in statistical models. Missing data can lead to incorrect conclusions and affect the performance of machine learning algorithms if not addressed properly.

2. What methods can be used to handle missing values in a dataset?

Common methods to handle missing values include:

1. Dropping rows with missing values.
2. Filling missing values with a specific constant (e.g., 0).
3. Replacing missing values with statistical measures like the mean, median, or mode of the column.
4. Using interpolation or predictive models to estimate missing values.

3. In the provided dataset, how would you fill the missing values using the mean of each column?

To fill missing values with the mean, you can use the `fillna()` method in pandas. For example, you can calculate the mean for each column and replace missing values with those calculated means: `df_filled_mean = df.fillna(df.mean())` This replaces any NaN values in the DataFrame with the mean of the respective columns.

4. What impact does dropping rows with missing values have on the analysis?

Dropping rows with missing values can lead to a reduced dataset, which might result in a loss of valuable information. If many rows contain missing values, this method could skew the analysis and reduce statistical power. However, it can be useful if the remaining data is sufficient for reliable analysis or if the missing data is assumed to be random.

5. How might filling missing values with zeros affect the interpretation of the data?

Filling missing values with zeros may artificially inflate the total consumption readings and misrepresent actual power usage. This can lead to incorrect conclusions about usage patterns and trends, especially if zeros do not accurately reflect the reality of the missing data. It's important to consider the context of the data and choose an appropriate method for filling missing values to ensure meaningful analysis.

RESULT:

Thus, the Python program successfully identified the missing values in the given dataset and handled them using appropriate data-cleaning techniques. The missing values were replaced or removed as required, and the cleaned dataset was displayed and verified successfully.

INFERENCE:

The experiment demonstrates that Python, particularly Pandas, provides effective methods for identifying and handling missing data. Proper treatment of missing values improves data quality, consistency, and reliability, making the dataset suitable for further analysis and decision-making.

Assessment	Max. Marks	Marks Obtained
Conduction of Practical	15	
Report	5	
Viva	5	
Total Marks	25	
Signature of Lab In-Charge		